

Homework Assignment

1) Clustering Task

Using the dataset **brich3.txt**, apply an appropriate clustering algorithm of your choice (e.g., K-means, hierarchical clustering, DBSCAN, etc.) to identify meaningful groups within the data.

Your report should include:

- A brief description of the selected algorithm and justification for your choice
- Data preprocessing steps (if any)
- Selection of model parameters
- Visualization of the clustering results

2) Market Basket Analysis and Business Recommendations

Using the datasets **bread basket.csv** and **Groceries data.csv**, perform an association rule analysis to uncover relationships between products.

Based on your findings, provide practical recommendations to shop owners regarding:

- Product placement and shelf organization

Your recommendations should be supported by quantitative metrics (e.g., support, confidence, lift) and clearly explained in terms of how they could increase sales.